$$v_3' = ev_3 \tag{17}$$

$$E_{k3}' = e^3 E_{k3} \tag{18}$$

Before its fourth collision with a plate, the ball has kinetic energy

$$E_{k4} = E_{k3}' + qU \tag{19}$$

From (3)、(15)、(18)、(19),

$$E_{k4} = \left(1 + e^2 + e^4\right) C_0 U^2 \tag{20}$$

Its speed just before the fourth collision is

$$v_4 = \sqrt{\frac{2E_{k4}}{m}} \tag{21}$$

Its speed and kinetic energy just after the fourth collision are

$$v_4' = ev_4 \tag{22}$$

$$E_{k4}' = e^2 E_{k4} \tag{23}$$

Continuing likewise, its kinetic energies just before and after collision n are

$$E_{kn} = \left[1 + e^2 + \cdots + e^{2(n-2)}\right] C_0 U^2 \tag{24}$$

$$E_{kn}' = e^2 \left[1 + e^2 \cdots + e^{2(n-2)}\right] C_0 U^{22} \tag{25}$$

Thus

$$E_{kn} = \frac{1 - e^{2(n-1)}}{1 - e^2} C_0 U^2 \tag{26}$$

$$E_{ln}' = \frac{e^2\left[1 - e^{2(n-1)}\right]}{1 - e^2} C_0 U^2 \tag{27}$$

For inelastic collisions, $e < 1$; E_{kn} and E_{kn}' grow with the collision count. As $n \to \infty$,

$$E_{k?} = \frac{1}{1 - e^2} C_0 U^2 \tag{28}$$

$$E_{k\infty}' = \frac{e^2}{1 - e^2} C_0 U^2 \tag{29}$$

The maximum kinetic energy is the limiting value just before a collision, so

$$E_{kmax} = E_{k\infty} = \frac{1}{1 - e^2} C_0 U^2 \tag{30}$$

2. After a sufficiently long time, as $n \to \infty$, the speeds just after one collision and just before the next are

$$v_\infty' = \sqrt{\frac{2E_{k\infty}'}{m}} = eU\sqrt{\frac{2C_0}{\left(1 - e^2\right)m}} \tag{31}$$